Angle between two planes

Basic skills

Q1. a) $2x - 3y + 5z = 10$

$\underline{n} = (2, -3, 5)$

$ax + by + cz = d$
normal
(a, b, c)

b) $2y + z = 10 + 2x$

so $-2x + 2y + z = 10$

$\underline{n} = (-2, 2, 1)$

c) $\left(\underline{r} - \begin{pmatrix} 2 \\ 5 \\ 0 \end{pmatrix}\right) \cdot \begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix} = 0$

form
$(\underline{r} - \underline{a}) \cdot \underline{n} = 0$

so $\underline{n} = \begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix}$

d) $(0,0,0)$, $(1,2,3)$, $(-1,-5,2)$

so two in-planar vectors are

$\underline{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\underline{b} = \begin{pmatrix} -1 \\ -5 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ -5 \\ 2 \end{pmatrix}$

then

$\underline{n} = \underline{a} \times \underline{b} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 2 & 3 \\ -1 & -5 & 2 \end{vmatrix} = \underline{i} \begin{vmatrix} 2 & 3 \\ -5 & 2 \end{vmatrix} - \underline{j} \begin{vmatrix} 1 & 3 \\ -1 & 2 \end{vmatrix} + \underline{k} \begin{vmatrix} 1 & 2 \\ -1 & -5 \end{vmatrix}$

$= 19\underline{i} - 5\underline{j} - 3\underline{k}$

e) $\underline{r} = \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}$ $\underline{r} = \begin{pmatrix} 2 \\ 0 \\ 6 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 5 \end{pmatrix}$

$\underline{d}_1 = \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}$ $\underline{d}_2 = \begin{pmatrix} 1 \\ 0 \\ 5 \end{pmatrix}$ ae in-planar vectors

$$\underline{n} = \underline{d}_1 \times \underline{d}_2 = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 0 & 2 & -1 \\ 1 & 0 & 5 \end{vmatrix} = \underline{i} \begin{vmatrix} 2 & -1 \\ 0 & 5 \end{vmatrix} - \underline{j} \begin{vmatrix} 0 & -1 \\ 1 & 5 \end{vmatrix} + \underline{k} \begin{vmatrix} 0 & 2 \\ 1 & 0 \end{vmatrix}$$

$$= 10\underline{i} - \underline{j} - 2\underline{k}$$

Q2. $\underline{a} = \begin{pmatrix} 4 \\ 1 \\ -3 \end{pmatrix}$ $\underline{b} = \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix}$

Angle by $\cos\theta = \dfrac{\underline{a} \cdot \underline{b}}{|\underline{a}||\underline{b}|}$

$$\cos\theta = \frac{20 - 1 - 6}{\sqrt{4^2+1^2+3^2}\sqrt{5^2+1^2+2^2}} = \frac{13}{2\sqrt{195}} \qquad \theta = 62.3^\circ$$

Competency skills

Q3. $\Pi_1: 2x - 3y - z = 9$ $\Pi_2: x + 6y + 2z = 20$

a) $\underline{n}_1 = (2, -3, -1)$ $\underline{n}_2 = (1, 6, 2)$

b) $$\cos\theta = \frac{\underline{n}_1 \cdot \underline{n}_2}{|\underline{n}_1||\underline{n}_2|} = \frac{2 - 18 - 2}{\sqrt{2^2+3^2+1^2}\sqrt{1^2+6^2+2^2}} = \frac{-18}{\sqrt{574}} \qquad \theta = 138.7^\circ \text{ or } 41.3^\circ$$

angle between normals

for acute $180 - \theta$

Q4. $x+y+z=4$, $x+2y-z=5$

$\underline{n}_1=(1,1,1)$ $\quad \underline{n}_2=(1,2,-1)$

then $\cos\theta=\dfrac{\underline{n}_1\cdot\underline{n}_2}{|\underline{n}_1||\underline{n}_2|}=\dfrac{1+2-1}{\sqrt{3}\sqrt{6}}=\dfrac{2}{3\sqrt{2}}$ $\quad \theta=61.9^\circ$

Q5. π_1 has in-planar vector $\begin{pmatrix}3\\-1\\2\end{pmatrix}$

and points $(0,0,0)$, $(5,1,3)$

π_2 has in-planar vector $\begin{pmatrix}3\\-1\\2\end{pmatrix}$

and points $(0,1,1)$, $(5,1,3)$

a) For π_1: has in-planar vector

$$\begin{pmatrix}5\\1\\3\end{pmatrix}-\begin{pmatrix}0\\0\\0\end{pmatrix}=\begin{pmatrix}5\\1\\3\end{pmatrix}$$

so normal

$$\underline{n}_1=\begin{pmatrix}3\\-1\\2\end{pmatrix}\times\begin{pmatrix}5\\1\\3\end{pmatrix}=\begin{vmatrix}\underline{i} & \underline{j} & \underline{k}\\ 3 & -1 & 2\\ 5 & 1 & 3\end{vmatrix}=-5\underline{i}+\underline{j}+8\underline{k}=\begin{pmatrix}-5\\1\\8\end{pmatrix}$$

For π_2: has in planar vector $\begin{pmatrix}5\\1\\3\end{pmatrix}-\begin{pmatrix}0\\1\\1\end{pmatrix}=\begin{pmatrix}5\\0\\2\end{pmatrix}$

so normal

$$\underline{n}_2=\begin{pmatrix}3\\-1\\2\end{pmatrix}\times\begin{pmatrix}5\\0\\2\end{pmatrix}=\begin{vmatrix}\underline{i} & \underline{j} & \underline{k}\\ 3 & -1 & 2\\ 5 & 0 & 2\end{vmatrix}=-2\underline{i}+4\underline{j}+5\underline{k}=\begin{pmatrix}-2\\4\\5\end{pmatrix}$$

b) so for angle

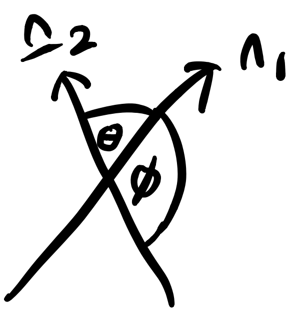
$$\cos\theta = \frac{\underline{n}_1 \cdot \underline{n}_2}{|\underline{n}_1||\underline{n}_2|} = \frac{10+4+40}{\sqrt{5^2+1^2+8^2}\sqrt{2^2+4^2+5^2}} = \frac{54}{45\sqrt{2}}$$

$$\theta = 31.9^\circ$$

Advanced skills

Q6. Intuatively it shouldn't as otherwise multiple angles are equivalent.

However there are two angles between planes



with $\theta + \phi = 180^\circ$

one acute
one obtuse

May get obtuse or acute depending on direction of $\underline{n}_1, \underline{n}_2$. Scale does not affect it.

As $\cos\theta = -a$ and $\cos\phi = a$

Q7. $x+y+z=0$, $x+y-kz=0$

a) solving by elimating

$$z - -kz = 0 \quad \text{so} \quad (k+1)z = 0$$

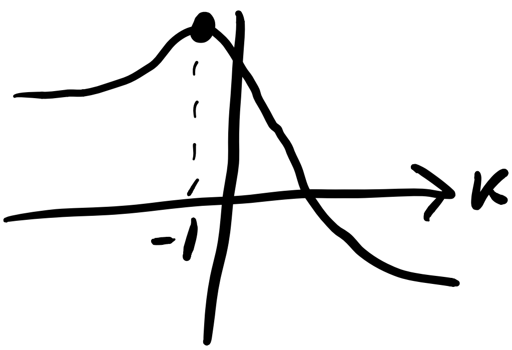
so $z=0$ means intersection and

then any point $x+y=0$, $z=0$
a line

b) $\underline{n}_1 = (1,1,1)$ $\quad \underline{n}_2 = (1,1,-k)$

$$\cos\theta = \frac{1+1-k}{\sqrt{3}\sqrt{2+k^2}} = \frac{2-k}{\sqrt{6+3k^2}}$$

For minimal angle want $\cos\theta = 1$
or as close as possible, so max
of $\dfrac{2-k}{\sqrt{6+3k^2}}$



derivative $\dfrac{-2(k+1)}{\sqrt{3}(2+k^2)^{\frac{3}{2}}}$ so at $k+1=0$ $k=-1$
for max, so min angle

c) For $k=1$

$$\cos\theta = \frac{2-1}{\sqrt{0+3}} = \frac{1}{\sqrt{9}} = \frac{1}{3} \qquad \theta = 70.5^\circ$$

d)

$x+y+z=0$ intersects at

$x+y+2=0$

$y=-x-2$

$z=2$

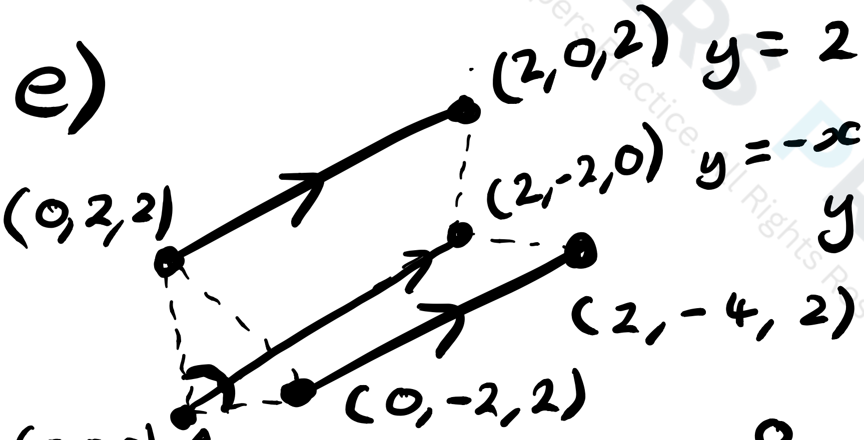
$x+y-z=0$ intersects at

$x+y-2=0$

$y=2-x$

$z=2$

e)



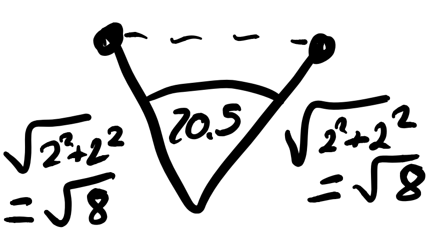
$y=2-x \quad (z=2)$

$y=-x \quad (z=0)$

$y=-x-2 \quad (z=2)$

angle is 70.5°

Triangle



so area is $\frac{1}{2}\sqrt{8}\sqrt{8}\sin 70.5$

$= 32\sin 70.5 \simeq 30.16$

Length

$\sqrt{2^2+2^2+0^2} = \sqrt{8}$

so volume is $30.16 \times \sqrt{8}$

$= 85.3$